

The Capstone Project Complete Guide

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Introduction and General Tips

1. What is the Capstone Project?

The **Capstone Project** is a part of the school's curriculum. It is one semester long; you must do a **Capstone Project** every semester. Students collaborate in small groups and go through the **Engineering Design Process (EDP)** to solve real-life problems related to **Egypt's Grand Challenges** (learn about the [EDP](#)). The **Capstone Project** makes up **60%** of all school subjects, and it is further divided into four categories: **Personal Journals** (40%), **Prototype** (10%), **Capstone Portfolio** (10%), and **Capstone Exhibition Poster** (40%), as shown in Table 1. **Personal Journals** are assessed by teachers using the [Personal Journal Rubric](#), and the three other categories are assessed by external evaluators using the [Capstone Exhibition Rubric](#).

Category	Assessment Tool	Evidence
Personal Journals - 40%	Personal Journal Rubric	Highest four of the five Student Personal Journals that are administered biweekly in the Capstone session
Prototype - 10%	Capstone Exhibition Rubric	Capstone Prototype
Capstone Portfolio - 10%	Capstone Exhibition Rubric	Capstone Portfolio
Capstone Exhibition Poster - 40%	Capstone Exhibition Rubric	Capstone Exhibition Poster

Table 1: Assessment and grading of the Capstone Project

2. What are Egypt's Grand Challenges?

Egypt's Grand Challenges (GCs) are the 11 most threatening challenges to Egypt's development. Any of them are "too grand" to be solved by one group project; hence, students must aim to solve a specific sub-problem of the **GCs** (known as the [Problem to Be Solved](#)). The **GCs** help with finding and defining the **Problem to Be Solved**, as it is closely related to more than one **GC**. **Egypt's Grand Challenges** are

- i. Improve the use of alternative energies
- ii. Recycle garbage and waste for economic and environmental purposes
- iii. Deal with urban congestion and its consequences
- iv. Work to eradicate public health issues/disease
- v. Increase the industrial and agriculture bases of Egypt
- vi. Address and reduce pollution fouling our air, water and soil
- vii. Improve uses of arid areas
- viii. Manage and increase the sources of clean water
- ix. Deal with population growth and its consequences

- x. Improve the scientific and technological environment for all
- xi. Reduce and adapt to the effect of climate change

3. What is the Capstone Portfolio?

The **Capstone Portfolio** represents the biography of the **Capstone Project** organized around the **EDP** components. It is used as a source of information to prepare for the **Capstone Exhibition Poster** and contribute to the **Capstone Exhibition**. Using **Microsoft Word**, one of the capstone team members sets up the **Capstone Portfolio** following the [Capstone Portfolio Template](#) and ensures that all team members have access to it; you can use **Microsoft OneDrive** for uploading and sharing the **Capstone Portfolio**. (You can use this complete guide or the **Capstone Portfolio Template**; both will guide you well.)

4. What is the Capstone Exhibition Poster?

The **Capstone Exhibition Poster** is the primary tool used to communicate the project's progress, research, scientific base, and results to the external evaluators during the **Capstone Exhibition**. Being a scientific poster, it is often regarded as the "summary" of the entire **Capstone Project**. It consists of the **Capstone Portfolio's Chapter III** and **Chapter IV**. However, some parts of **Chapter III** and **Chapter IV** are not included in the **Capstone Exhibition Poster**; read [Chapter III](#) and [Chapter IV](#) to learn what parts are in the **Capstone Exhibition Poster**. One of the **Capstone Exhibition Poster's** requirements is to be clear and look appealing; you will learn more about this in the [Poster Layout](#) section.

5. General Tips

Throughout writing the **Capstone Portfolio** and **Capstone Exhibition Poster**, some tips will help you:

- i. Correctly format your **Capstone Portfolio** Microsoft Word document:
 - a. Follow the [Capstone Portfolio Template](#) and include a **Table of Contents** at the start.
 - b. Justify (as in "to click **Ctrl+J**" and not "to give reasons for") all body paragraphs.
 - c. Use **Times New Roman** as the primary font for the **Capstone Portfolio** and **Capstone Exhibition Poster**.
 - d. Set the body font size to 14 points in the **Capstone Portfolio**.
 - e. See this [Capstone Portfolio Formatting Guide](#).
- ii. Avoid using any personal pronouns (e.g., I, you, he, she, we, us, him, her, his, hers, its, theirs, our, and your).
- iii. Use punctuation marks (e.g., semi-colons, colons, hyphens, en-dashes, and em-dashes) correctly.
- iv. Make use of online grammar checkers (see [Grammarly](#), which you can get as [an add-in for Microsoft Word](#)) to ensure the quality of your writing.
- v. Use the **American Psychology Association 7th edition (APA 7)** correctly (see this [guide](#)

or this [guide](#)). Visit and read the two linked guides. Also read the following tips that you might not notice/find from reading the guides:

- a. Visuals in **APA 7** are either figures or tables (yes, there are **no graphs**).
 - b. There are also non-visual equations, which you need to mention, too.
 - c. Have descriptive captions for all figures, tables, or equations. Always mention figures, tables, or equations in your text at least once. Also, [cite them](#).
 - d. When mentioning figures, tables, or equations, write Figure N, Table N, or Equation N with a capitalized F, T, or E, where "N" refers to the number of the figure, table, or equation.
 - e. In the figure, table, or equation captions, there are no periods (.) at the end (e.g., "Figure 1: Egypt's population growth from 1990 to 2012").
 - f. Make use of citation machines; make sure you choose APA 7 and review the generated citation before putting it in the **Capstone Portfolio** (see [Scribbr Citation Machine](#))
 - g. Do not remove the indent or unitalicize words in the bibliography (see this [journal article citation](#) example or this [book citation](#) example).
 - h. Use [in-text citations](#) before the sentences' periods (for sentences) or after the paragraphs' periods (for paragraphs) to direct the reader to the exact source of the information.
- vi. Do **NOT** plagiarise (see this [infographic](#)) and make sure your writing is original by using a plagiarism checker.
 - vii. Be clear, concise, descriptive, and cohesive. Also, avoid redundancy as much as possible.
 - viii. Use evidence like figures, tables, equations, facts, and statistics to prove, explain, or show points. Avoid solely relying on "logic" or "common sense."

I. Present and Justify a Problem and Solution Requirements

1. Egypt Grand Challenge(s)

Introduce Egypt's Grand Challenges, explain what they are, and explain why they are critical to address and solve.

In one small introductory paragraph, you should explain what Egypt's Grand Challenges are. List the 11 **Grand Challenges** **after clarifying why they greatly threaten Egypt's development** and impede its economy, agriculture, and industry. Write the paragraph under the title **"Introduction."** This section should be right before the first **Grand Challenge**.

Which Egyptian Grand Challenge(s) are you addressing, and why are they important to address?

Specify (four to five) **Grand Challenges** that are related to the **Capstone Problem** or big idea.

Each **Grand Challenge** consists of three sections: **Overview**, **Causes**, and **Impacts**. The "**Overview**" section answers the question above using evidence (statistics, figures, tables, etc.) and not just using logic or common sense (which applies to the **entirety** of the **Capstone Portfolio**). You can have multiple paragraphs, figures, or tables in the "**Overview**" section. The "**Causes**" and "**Impacts**" sections contain **three to four causes and three to four impacts**; each cause is a paragraph, and each impact is a paragraph. **Make sure to mention at least one piece of evidence for each cause or impact**. Avoid using multiple figures, tables, or equations in a single cause or impact.

Note that the **Causes** and **Impacts** sections should discuss **the causes of the Grand Challenge's emergence** and the negative impacts of the **Grand Challenge's** unstoppable existence. For example, the causes and impacts of the "**Improve the use of alternative energies**" **Grand Challenge** should discuss why there is a lack of alternative energies and emphasize **the negative impacts of having a lack of alternative energies**.

2. Problem to Be Solved

What specific problem are you addressing?

You answer this question in the first section of the **Problem to Be Solved**, the "**Overview**" section. Make sure that the **Problem to Be Solved** you chose is more specific than the **Egypt Grand Challenges** and is more specific than just connecting those **Grand Challenges**. You can use **multiple paragraphs in the "Overview" section**; however, **you should avoid writing figures, tables, or equations in it**. Try starting with the broad idea of two or three main **Egypt Grand Challenges** and then arriving at a more specific idea by mentioning the main connection between these **Grand Challenges**. Then, you would try to make this connection more specific to reach the **Problem to Be Solved**. Your goal is to achieve coherent interconnected paragraph ideas that would make up an essay for the **Problem to Be Solved's Overview** section.

What are the impacts or consequences related to dealing with that problem?

Explain the major positive (if the problem is solved) and negative (if the problem is not solved) consequences, listing three to four negative consequences (each in a paragraph) and three to four positive consequences (each in a paragraph). Hence, the **Problem to Be Solved** consists of three sections: **Overview**, **Positive Consequences (If Solved)**, and **Negative Consequences (If Not Solved)**. Avoid using multiple paragraphs, figures, or tables in a single consequence. Avoid redundancy: for example, do not mention "Reduced Greenhouse Emissions" as a positive consequence and then mention "Greenhouse Effect" as a negative consequence.

3. Research

Which topics did you research about the problems?

Explain different topics that will help in a deeper understanding of the problem. This section is called "**Topics Related to the Problem**." It consists of further researched topics that lead you to understand the specific problem better. They include but are not limited to the specific problem, its causes, impacts, and related aspects. They should be addressed as negative topics related to the continuity of the problem without being dealt with. One possible example is Photochemical smog's rise as an impact on the usage of building materials that are not environmentally safe (where their industrial processes have greenhouse gas emissions) to build pedestrian tunnels. Lastly, the section consists of three to four topics addressed according to an Egyptian perspective (avoid broad definitions and

statistics). The three or four topics can contain multiple paragraphs, figures, or tables.

Which topics did you research about the solutions?

Explain different topics related to the possible approaches to solving the problem. This section is called "**Topics Related to the Solution.**" These topics are considered subtopics of your solution. These topics supposedly helped you better understand and apply your solution. They should be related to the chosen solution. Examples of these topics are tunnel construction techniques and scientific concepts related to the solution. For instance, if the **Capstone Challenge** is to build a tunnel, the modulus of elasticity, stress, and strain are physics concepts that will help in choosing reinforcement bars materials for the tunnel.

4. Other Solutions Already Tried

What prior solutions have been applied to try to solve this problem?

Mention and explain four past solutions you have found when researching how you can solve this problem yourself. These solutions should be actual engineering projects that had the goal of attempting to solve the problem and not just projects that can be interpreted as a prior solution to the problem. In each of the four prior solutions, you have four sections: **Overview**, **Mechanism**, **Points of Strengths**, and **Points of Weaknesses**. In the **Overview** section, you will answer, "What prior solutions have been applied?" Mention essential details: **who is responsible for the solution**; where **the solution is based**; **when the solution was built**; and what problem the solution was exactly trying to solve.

How do these prior solutions work? What are their mechanisms?

In the **Mechanism** section, you will answer this question in detail. Utilize figures to explain better how each solution works.

What are the strengths and weaknesses of those solutions?

In the **Points of Strengths** and **Points of Weaknesses** sections, you will answer this question using three or four points of strengths and three or four points of weakness. Avoid using multiple paragraphs, figures, or tables for each point of strength or weakness.

II. Generating and Defending a Solution

1. Solution and Design Requirements

What characteristics (Solution Requirements) should a successful solution have, and how each of these characteristics could make the solution successful?

- 1 In the first "**Solution and Design Requirements**" section, **Solution Requirements**, you answer this **question with general solution requirements** that must be in any **effective, efficient, and successful solution**. Examples of solution requirements are availability, durability, eco-friendliness, and cost-effectiveness. **Each solution requirement consists of one paragraph and no figures or tables.**

Which Design Requirements did you choose, and why did you choose them?

- 2 In the second "**Solution and Design Requirements**" section, **Design Requirements**, you answer

this question with design requirements that **resemble your prototype's benchmark** of acceptance and rejection, where there should be one or more distinct aspects that will test your prototype's effectiveness to be considered applicable in real-life projects. The design requirements are predetermined in the first semester of your sophomore year, but later on, you should choose your design requirement(s) and justify why you chose them. Also, explain how each design requirement relates to the solution. Remember that a design requirement must be measurable. This section (**Design Requirements**) should be only one paragraph that includes an ending sentence referring to how your prototype will attain these requirements after being tested.

2. Selection of Solution

another paragraph about material
another paragraph real life application

Describe the solution you decided to pursue and why you chose it.

Explain the solution in detail using multiple paragraphs. Remember that in this section, you demonstrate the idea of the prototype you will build to solve the problem and not the prototype itself.

3. Selection of Prototype

Describe in detail the prototype (design and construction) you have decided to construct and how it will meet the Design Requirements that you have chosen.

Add a SketchUP 3D design as a Figure and describe your prototype in detail. Try to clearly show why you have chosen the dimensions of your prototype; justify all dimensions. Explain the methodology of your prototype in detail. Note that 3D designing using another computer program other than SketchUP is acceptable.

III. Constructing and Testing a Prototype

1. Abstract (Not Included In the Capstone Portfolio)

The **Abstract** is a brief **150–250 words** paragraph describing the work described in the poster. The **Abstract** must give the reader a clear understanding of the project without reading the entire poster. Tie the entire project together by summarizing the two or three main **Grand Challenges**, the **Problem to Be Solved**, **chosen solution**, **Design Requirements**, the **prototype**, **Results**, and **Conclusions**. The **Abstract** includes (1) the **purpose of the study**, (2) a brief statement of **what was done** (without including minor details of the methods), (3) a brief statement of significant **findings**, and (4) major **conclusions**. The **Abstract** should generate excitement and desire to learn more about the topic; put great effort into writing it. Lastly, it does not include any visuals.

2. Introduction (Not Included In the Capstone Portfolio)

In the **250–400 words Introduction** paragraph, a connection is made between two or three main **Grand Challenges**, the semester's **Capstone Challenge**, the **Problem to Be Solved**, **two prior solutions** (mentioning their **strengths** and **weakness**), **chosen solution** (and **how it was chosen**), and the **Design Requirements**. The Introduction must include a figure that highlights/explains the significance or impact of the two or three main **Grand Challenges** or the **Problem to Be Solved**. The **Introduction** should move clearly from a broad view of the **Grand Challenges** to an increasingly narrow focus on the team's chosen solution, **Design Requirements**, and **prototype**, justifying each choice (see the [Funnel Technique](#)). It should also provide smooth transitions from "what" choices were made to "why" they were made to the next section. At the end of the paragraph,

you must transition to the next section, **Materials and Methods**, by mentioning that certain materials and methods were used to construct and test the prototype.

3. Materials and Methods

What materials did you use to build the prototype?

Add a table that contains the **Item** (as in the name), **Quantity**, **Description**, **Usage**, **Cost**, **Source of Purchase**, and **Picture** of all the items you have used in your prototype. Remember to add the correct measuring units in the **Quantity** column. Do not forget to add a caption to this table and mention it. This table is included in the **Capstone Exhibition Poster**; however, the table would only include the **Item**, **Quantity**, **Description**, and **Usage** of the items. Remember to check the **Capstone Challenge** to ensure the prototype does not exceed the maximum cost (typically 1,000 Egyptian Pounds).

This part **IS INCLUDED** in the **Capstone Exhibition Poster** under the "Materials and Methods" title.

How did you construct this prototype using the materials mentioned above?

This section is often referred to as the "**Methods**" section; however, both **Materials** and **Methods** are inseparable as sections but rather one section named **Materials and Methods**. Mention the "**Materials**" table (e.g., "As shown in Table N, the materials were used to...") and explain all the construction steps you took to build the prototype. Include Figures of the most crucial construction steps; that is why you should always document your construction using images and videos.

This part **IS INCLUDED** in the **Capstone Exhibition Poster** under the "Materials and Methods" title.

List the safety precautions you took to ensure your team is being safe.

Moving on from the construction steps, you should write a separate paragraph about all the safety precautions your team took to ensure safety and prevent danger. These precautions include but are not limited to wearing protective clothing. Mention all types of protective clothing you have used.

This part **IS NOT INCLUDED** in the **Capstone Exhibition Poster**.

4. Test Plan

List the Design Requirements you have chosen to test with your prototype.

This is considered a summary of the **Design Requirements** section in **Chapter II's Solution and Design Requirements**. Do not mention any justification or reasoning here.

This part **IS INCLUDED** in the **Capstone Exhibition Poster**.

List the steps for each test plan on the prototype associated with each design requirement.

This is the **Test plan** part; it is very similar to the beforementioned "construction steps." Explain all the steps you took to conduct the test plan. Like the construction steps, you should include figures of the most crucial test plan steps, so always document your test plan steps using images and videos. Address how the test plan addresses the beforementioned **Design Requirements**. During the **Capstone Exhibition**, you will be asked to show your test plan video and the images, so document your test plan(s).

This part (in addition to the **Design Requirements** part) **IS INCLUDED** in the **Capstone Exhibition Poster**; however, both the **Test Plan** and **Design Requirements** parts are parts under the "**Materials and Methods**" title.

5. Data Collection

What measurement tools or instruments did you use? What level of precision, as well as the error, did you use in measurement?

These tools or instruments include but are not limited to sensitive balance, vernier caliper, and voltmeter. There is no limit to how many tools or instruments you can include. Write only one paragraph that answers these two questions together.

This part **IS NOT INCLUDED** in the **Capstone Exhibition Poster**.

List all data collected in each test plan.

This part has two sections: **Negative Results** and **Positive Results**. In the **Negative Results**, you explain all the failed test plan trials and their reasoning. In the **Positive Results** section, you explain the positive thoroughly. Use visual data representation using tables or graphs. Also, avoid pie charts, which are often misleading (as angles are not easily estimated and compared by humans). Remember to include the correct measurement units and quantities for the inaccuracies/errors ($\pm$). The visual representation of the results alone (without the words) should lead the reader to a conclusion about the results. The results should be authentic and replicable.

This part **IS INCLUDED** in the **Capstone Exhibition Poster** but with a different title: "**Results**."

IV. Evaluation, Reflection, Recommendations and Discussion

1. Analysis and Discussion

What is the analysis of the prototype behavior? What is the analysis of the test plan results?

You answer this question by dissecting it into different ideas, with each consisting of a paragraph or figure. Tie performance results (positive and negative) to the original challenge or problem being addressed and to the **Grand Challenge**. Tie the prototype testing results to the **Design Requirements**. The prototype test results, pictures, graphs, charts, and other visuals must support the analysis. The analysis must be supported by scientific laws and theories. Also, it should provide evidence of learning transfer of discipline **Learning Outcomes (LOs)**. It must be clear, organized, and well-developed. It should use explanation and justification to persuade the reader that you have a good understanding and strong scientific base for the whole project. It is written to meet the needs of the intended audience.

This part **IS INCLUDED** in the **Capstone Exhibition Poster** under the title "**Analysis**."

What conclusions do you reach from the data you collected?

The test and measurements should be accurate enough to draw conclusions. The data should authenticate that the prototype met the identified **Design Requirements**. Conclusions must be drawn from the prototype test results and analysis. Conclusions drawn from prototype test results and

analysis should be compared with the team's research on other solutions.

This part **IS INCLUDED** in the **Capstone Exhibition Poster** under the title "**Conclusions.**"

2. Recommendations

What recommendations do you have for future work in this area?

This includes recommendations on the prototype and the project idea's real-life application. Recommendations must be specific; mention ways the project could be improved in the future. Recommendations should be practical and directed towards a future research, engineering, or policy group. Recommendations should clearly be informed by the problem and their proposed solutions. Avoid recommending something that you could have done. With each recommendation (a way that the project could be improved), you should explain why you were not able to achieve this improvement (e.g., the price was high, it was unsafe, the required tools were not available, etc.).

This part **IS INCLUDED** in the **Capstone Exhibition Poster** under the title "**Recommendations.**"

Tell another team, who wanted to start where you stopped on your solution, what they should work on and consider improving.

This part is written as a small paragraph using active voice and personal pronouns (such as I, we, etc.) with no figures or tables. Include this in the **Capstone Portfolio** under the title "**Recommendations to Other Teams**" or "**Recommendations to Future Teams.**"

This part **IS NOT INCLUDED** in the **Capstone Exhibition Poster**.

How did working on this project help you and your team to become better STEM School students?

This part includes the scientific, engineering, and social impacts. Include this in the **Capstone Portfolio** under the title "**Project's Positive Influence**" or "**Project's Benefits.**"

This part **IS NOT INCLUDED** in the **Capstone Exhibition Poster**.

3. Learning Outcomes

Which ten discipline learning outcomes did you identify as related to your Design Challenge? Explain how each identified learning outcome was transferred to your Capstone.

Add a table containing ten rows (for ten discipline **Learning Outcomes**) and four columns: **Learning Outcome (LO)**, **Concept**, **Content**, and **Relation**. When writing an **LO** in the **LO column**, first write the name of the subject and the **LO** code in brackets: **Physics (PH.1.01)**. Remember that each subject has respective initial letters in its codes: Math is **MA**, Mechanics is **ME**, Earth Science is **ES**, Chemistry is **CH**, Biology is **BI**, and Technology is **CS**. The **Relation column** must contain a paragraph that explains how the **LO** was helpful during the **Capstone Project**. Do not forget to add a caption to this table.

This part **IS NOT INCLUDED** in the **Capstone Exhibition Poster**.

4. List of Sources in APA Format

List all research sources you used while writing the **Capstone Portfolio** using APA 7 format.

This part **IS INCLUDED** in the **Capstone Exhibition Poster** under the title "**Literature Cited**." However, in the **Capstone Exhibition Poster**, you should only include five to seven (rather than all) credible and peer-reviewed sources (papers, books, or encyclopedias) that were used as references to the **Capstone Exhibition Poster** (or the **Capstone Portfolio's Chapter III** and **Chapter IV**).

Capstone Exhibition Tips

1. Poster Layout

When designing the **Capstone Exhibition Poster**, there are things to look out for to ensure the poster looks pleasing:

- i. Body text must be readable from a distance of one meter. I found **20.5–22.5 points** the best fit. Use this [Formula](#) to determine what works best for you and your team.
- ii. The project title should be at the top of the poster, short, descriptive of the project, and easily readable at about two meters (words about 1.5–2.5 cm tall or **43–71 points**). The title must not be just an acronym; however, adding an acronym next to the full title is acceptable.
- iii. The poster must include the presenters' and school's names in a section that is 20-30% smaller than the title (or **30–57 points**).
- iv. The poster must include the grade, semester, and academic year (e.g., **Grade 2, Semester 2, 2022–2023**).
- v. Images are in an appropriate size relative to the body text. The images should be clear from a distance of about one meter.

2. Presentation

When presenting during the **Capstone Exhibition**, there are things to look out for to make sure the presentation goes well:

- i. The **Capstone team** should present the entirety of the poster in 15 minutes; the **Capstone Exhibition** is divided into 15 minutes of presenting and 5 minutes of answering the evaluator's questions. However, if the evaluator wished to ask during the first 15 minutes, the team should use the entire 20 minutes to present and answer questions mid-presentation.
- ii. Starting from the **Introduction**, you should introduce yourself and your team.
- iii. Show that the materials are student-purchased, the prototype is student-built, and safety precautions were followed.
- iv. Respond to all of the evaluator's questions, comments, or requests; do not make them wait.
- v. Use formal, professional, and appropriate language. Never speak in Arabic unless you are directly asked to do so.
- vi. Speak with appropriate loudness and speed. Avoid speaking with the same tone by changing your tone when you want to emphasize key points.

- vii. Use gestures and body language. Demonstrate enthusiasm for the subject and show interest in evaluation. In short, act happy and energized.
- viii. Provide materials and handouts.
- ix. Have your test plan video(s) ready to roll.
- x. Have all your **Literature Cited** open and highlight everything (facts, statistics, figures, tables, equations, etc.) that you have used in your poster in preparation to answer any "Where did you get this information from" evaluator questions.
- xi. Have confidence stay respectful, and never raise your voice disrespectfully.
- xii. Be ready to start the test plan on command. If the test plan is too long, be ready to at least show parts of it, and make sure you explain why the entirety of the test plan would not take less than 20 minutes.

Personal Journals (Not Included In the Capstone Portfolio)

1. Overview

When are the Personal Journals (Journal Assessments) done?

Every two Mondays (or Wednesdays), all STEM School students of a certain grade (10, 11, or 12) sit in their **Capstone sessions** and answer the Journal's questions. During the 60 minutes, all students are proctored by their school's teachers. **Google Forms** is (often) used to collect the answers; sign in using your school-issued email (the one ending with "@stemoctober.moe.edu.eg").

How many Personal Journals are there?

There are five primary **Personal Journals** per semester. Additionally, a sixth **Personal Journal Makeup** is held at the end of the semester for all students that missed a primary **Personal Journal** due to some excuse.

How many Personal Journals are used to calculate the Capstone Grade?

Only the best four of five **Personal Journals** are considered when evaluating the **Personal Journals'** grade, which is 40% of the **Capstone Project** total grade.

What is the number of questions in a Personal Journal?

For Grade 10 students, there are only three questions. For Grade 11 and Grade 12 students, there are four questions.

How many words must be written in each prompt?

You must write **150–250 words** as an answer to each paragraph. Writing **less than 150 words** might result in a **Red** or **Yellow** score, while writing **more than 250 words** may result in an inability to submit your response.

2. Prompts' Categories

There are four main categories for the questions/prompts in **Personal Journals**: **Personal**

Reflection; Team Collaboration; Using the Engineering Design Process; and Learning Transfer.

- i. **Personal Reflection** questions ask about your viewpoints, interpretations, and personal growth as a person, STEM Student, **Capstone team member**, or **Capstone team leader**.
- ii. **Team Collaboration** questions ask about your role in the team setting and your opinions about how collaborative behavior would solve conflicts and problems. **Personal Reflection** and **Team Collaboration** are sometimes merged or interchanged, as they both inquire about personal aspects and opinions. The first prompt is always a **Personal Reflection** or **Team Collaboration** question.
- iii. **Using the Engineering Design Process** questions ask about your progress doing the **Capstone Project**; the **EDP** steps you have taken; and justification for your choices (for materials or solutions). Questions of this category require you to provide a clear, insightful reflection. To know if your reflection is good, ask yourself this question: "Is this reflection well-developed enough that others attempting the same or similar project would find it useful?" The second prompt is always a **Using the Engineering Design Process** question.
- iv. **Learning Transfer** questions require a basic understanding of the **Learning Transfers (LOs)** of the related-to-Capstone subjects in your curriculum. They ask how a specific concept, idea, or notion in an **LO** might help advance or help you in solving the problem. The connection between the **Capstone Challenge** and **LO** is often unclear at first glance; use your creativity to answer questions in this category. The third and fourth prompts are always **Learning Transfer** questions. Note that Grade 10 students do not have a fourth prompt.

3. Grading System

Answers are evaluated based on a color-based grading system. Each answer is evaluated and given one of four colors: **Red**, **Yellow**, **Green**, and **Blue**.

- i. A **Red** score indicates that the response did not answer the prompt or that it did not display a correct understanding of the topic being addressed. A **Red** score is **rare**. Note that plagiarizing will result in getting a **Red** score.
- ii. A **Yellow** score indicates that the response does not address every aspect of the prompt question. Questions have multiple parts, and *all* (not some) parts must be addressed. A **Yellow** score is **uncommon**.
- iii. A **Green** score indicates that the response addresses every aspect of the prompt question. If all the parts of the prompt are answered accurately, a **Green** score is given. A **Green** score means that the response showed a very comprehensive and complete understanding of the topics in the prompt. A **Green** score is the **most common**.
- iv. A **Blue** score indicates that the response meets the **Green** criteria and demonstrates an even deeper understanding. A **Blue** score is the highest score; it is given when you "go above and beyond" by showing creativity and mentioning specific, accurate examples in your response. A **Blue** score is **uncommon**.

4. Rules

Some mandatory-to-follow rules govern the **Journal Assessments**:

- i. Students must take the **Journal Assessment** during a **Capstone session**.
- ii. Students in the same grade at all schools must take the **Journal Assessment** simultaneously.
- iii. Students shall not work in groups but rather individually.
- iv. Students shall not use the internet to look up information.
- v. Students shall not use the **Capstone Portfolio** for information.
- vi. Students shall not use unauthorized prepared responses.
- vii. Students shall not use other students' answers.
- viii. Students shall not purposefully allow other students to see or use their answers.
- ix. Students with an excused absence are allowed one **Personal Journal Makeup**.
- x. Every student takes five **Journal Assessments**, and the lowest score is dropped.

5. Tips

Since 40% of your **Capstone Project**'s grade relies on your **Personal Journal** scores, you must ensure the quality of your responses. Here are a few tips:

- i. Although it is acceptable to only write 150 words, I personally advise you to write **200–250 words** in each answer. This is because, unfortunately, quantity matters more than quality in some circumstances, and the **Journal Assessment** is sometimes such a circumstance.
- ii. When answering **Personal Reflection** or **Team Collaboration** questions, your response should be about what you (singular) think and feel rather than the entire team. Avoid saying "we" or "us." Keep the answer about yourself.
- iii. Divide your answer into a brief introduction, detailed body paragraph, and summary conclusion. Ensure the main points you want to get across will be noticed.
- iv. Do not spend much of your time fixing grammar and punctuation mistakes. Poorly written, great, unique ideas will score higher than perfectly written generic, bad ideas.
- v. Use examples that help the reader understand your statements.
- vi. Use equations when answering **Learning Transfer** prompt. Mostly, there is at least one equation that is relevant.
- vii. The teacher who is going to assess your response is not an engineer or scientist. Do not assume that they will have a complete understanding of the **EDP** or any scientific concept you are using in your **Capstone Project**. You might imagine that a chemistry teacher would not need detailed explanations in your answer to a chemistry-related **Learning Transfer**, and that is true. However, all teachers grade your three (or four) answers, even if an **Learning Transfer** answer is unrelated to their subject. For example, a CS teacher can and will grade your answer to a chemistry-related **Learning Transfer** question, so ensure your answer is understandable by any teacher that teaches a completely unrelated subject.